

Status. THM048 and THM049 passed both fresh audit gates and root comparison. AUD069 reconstructed both claims; AUD070 reviewed them as a hostile referee. The original dynamic assertion THM046 remains open.

Work on the unit four-torus with $\widehat{g}(k) = |k|^{-2}$ for $k \neq 0$, $\widehat{g}(0) = 0$, $K = -\nabla g$, and $c = 4\pi^2$. The actual particles start from iid Haar and solve

$$dX_i = B_i dt + \sqrt{2\nu_N} dW_i, \quad B_i = \frac{1}{N} \sum_{j \neq i} K(X_i - X_j), \quad \nu_N = \beta_N^{-1}.$$

Assume $\lambda_N = \beta_N N^{-1/2} \rightarrow \lambda \in (0, \infty)$. All constants below are uniform on a fixed eventual positive bounded interval for λ_N , with fixed smooth real h and finite T . There is no prescribed rate or universal starting index. The old energy-floor condition in this row is $\beta_N \rightarrow 0$; full microscopic subcriticality is $\beta_N/\sqrt{N} \rightarrow 0$; neither is the present critical regime.

Let $f_t = Q_{T-t}^{\nu_N} h$, where Q_a^ν preserves constants and has multiplier $e^{-a(c+\nu|k|^2)}$ on nonzero modes. Put

$$J_t(x, y) = K(x - y) \cdot (\nabla f_t(x) - \nabla f_t(y)), \quad S_N = \sqrt{N} \int_0^T P_N[J_t] dt,$$

$$P_N[v] = \frac{1}{2N^2} \sum_{i \neq j} v(X_i, X_j) - \frac{1}{N} \sum_i \int v(X_i, y) dy + \frac{1}{2} \iint v.$$

Write $\rho_t = \eta_N(t) - dx$ and $\Delta_N = \sqrt{N}(\rho_T[h] - \rho_0[Q_T^{\nu_N} h])$.

Energy and two complementary bounds. For $H_N = N^{-1} \sum_{i < j} g(X_i - X_j)$, the complete force square and true energy martingale satisfy

$$H_N(X_t) + \int_0^t \sum_i |B_i|^2 da = H_N(X_0) + \nu_N c(N-1)t + E_t, \quad \langle E \rangle_t = 2\nu_N \int_0^t \sum_i |B_i|^2 da.$$

The heat-split floor is $H_N \geq -C\sqrt{N}$. Since $\mathbb{E}H_N(X_0) = 0$, localization, isometry and the maximal inequality give

$$\mathbb{E} \int_0^T \sum_i |B_i|^2 dt \leq C\sqrt{N} + \nu_N c(N-1)T, \quad \mathbb{E} \sup_{t \leq T} |E_t|^2 \leq C.$$

The deterministic source version of the R16 heat split, with every self and Haar term retained, gives $\sqrt{N}|P_N[J_t](x)| \leq C_h(G + H_N(x)/\sqrt{N})$. Therefore, for fixed $A, a \geq 0$,

$$|S_N| \leq A + \frac{a}{\sqrt{N}} (H_N(X_0)^+ + \sup_{t \leq T} |E_t|).$$

This bound alone cannot be squared: the initial energy has infinite second moment.

The exact one-body identity supplies the complementary estimate

$$S_N = \Delta_N - L_T, \quad L_t = \sqrt{\frac{2\nu_N}{N}} \sum_i \int_0^t \nabla f_a(X_i(a)) \cdot dW_i(a),$$

$$|\Delta_N| \leq 2\|h - \bar{h}\|_\infty \sqrt{N}, \quad \langle L \rangle_T \leq 2\nu_N T \|\nabla h\|_\infty^2.$$

All identities are genuine at each finite N , after collision localization. In particular the time-integrated source is square integrable even when the instantaneous source is not.

Initial clipping and squared uniform integrability. For $L \geq L_0$ let $m_L = \int \min(g, L)$ and $g^{[L]} = \min(g, L) - m_L$. The local Coulomb singularity gives

$$|\{g > L\}| \leq CL^{-2}, \quad -C/L \leq m_L \leq 0, \quad \|g^{[L]}\|_2^2 \leq C(1 + \log L).$$

At iid time zero, $H_N^{[L]} = N^{-1} \sum_{i < j} g^{[L]}(X_i - X_j)$ has

$$\mathbb{E}(H_N^{[L]})^2 = \frac{N-1}{2N} \|g^{[L]}\|_2^2.$$

Pairs sharing one label have zero covariance by conditioning on that label, not by independence. Set $L = L_0 N^2$ and let B_N be the initial event that some pair exceeds L . Then

$$\mathbb{P}(B_N) \leq CN^{-2}, \quad H_N(X_0) = H_N^{[L]} + \frac{N-1}{2} m_L \leq H_N^{[L]} \quad \text{on } B_N^c.$$

On B_N^c the energy bound costs $C(1 + \log N)/N$ in squared excess. On B_N the endpoint bound costs $CN\mathbb{P}(B_N)$, and initial measurability gives exactly

$$\mathbb{E}[\mathbf{1}_{B_N} |L_T|^2] = \mathbb{E}[\mathbf{1}_{B_N} \langle L \rangle_T] \leq C\nu_N \mathbb{P}(B_N).$$

This is conditional isometry, not independence from the flow. Together,

$$\mathbb{E}(|S_N| - A)_+^2 \leq C \frac{1 + \log N}{N} =: \varepsilon_N \rightarrow 0, \quad (\mathbb{E}|S_N|)^2 \leq \mathbb{E}|S_N|^2 \leq 2A\mathbb{E}|S_N| + 2\varepsilon_N.$$

The squared sources are uniformly integrable, including the finite initial segment. Since $\mathbb{E}|L_T|^2 = O(\nu_N) \rightarrow 0$, the squared endpoint defects are too, and

$$\mathbb{E}|S_N| \rightarrow 0 \iff \mathbb{E}|S_N|^2 \rightarrow 0 \iff \mathbb{E}|\Delta_N|^2 \rightarrow 0 \iff \mathbb{E}|\Delta_N| \rightarrow 0.$$

The original normalization equals $\sqrt{N}$ eventually. No assertion here proves any of these limits; the threshold A need not vanish.

The remaining signed-correlation criterion. For fixed $k \neq 0$, put $e_k = e^{2\pi i k \cdot x}$, $Z_k = \eta_N[e_k]$, $\ell_k = c|k|^2$, $a_k = c + \nu_N \ell_k$, $\tilde{a}_k = a_k - c/N$ and

$$j_k(x, y) = J^{e_k}(x, y) + c(e_k(x) + e_k(y)).$$

Let $A_2 = \mathbb{E}[j_k(X_1(t), X_2(t)) \overline{e_k(X_1(t))}]$ and define A_3 by replacing the last label by 3. Define B_2, B_3 by replacing only that last time by 0. Set

$$F_N = \frac{N-1}{N} A_2 + \frac{(N-1)(N-2)}{2N} A_3, \quad G_N = \frac{N-1}{N} B_2 + \frac{(N-1)(N-2)}{2N} B_3.$$

The three-label terms are absent at $N = 2$. All expectations use the actual law. Exact Itô and overlap counting give

$$N\mathbb{E}|Z_k(T) - e^{-a_k T} Z_k(0)|^2 = (e^{-\tilde{a}_k T} - e^{-a_k T})^2 + \frac{\nu_N \ell_k}{\tilde{a}_k} (1 - e^{-2\tilde{a}_k T}) + 2\Re \mathcal{R}_N,$$

$$\mathcal{R}_N = \int_0^T \left[e^{-2\tilde{a}_k(T-t)} F_N(t) - e^{-a_k T} e^{-\tilde{a}_k(T-t)} G_N(t) \right] dt.$$

The explicit errors are $O(N^{-2})$ and $O_k(N^{-1/2})$. With the new tail lemma and a polynomial L1 mode majorant, the original assertion for every fixed smooth real test is equivalent to $\Re \mathcal{R}_N(k, T) \rightarrow 0$ for each fixed mode. This signed cancellation remains open. A positive limsup would give an admitted fixed cosine or sine counterexample; no such witness is constructed.